FDISP

FRACTION STUDIO PROPORTIONAL TOWERS

4-Way Multi-Notational Fraction Blocks (10 Denominators: 1/1 to 1/12)

SERIES 1 • CARD 04 / 12 • DOMAIN: PROPORTIONAL TOWERS & MULTI-NOTATION EQUIVALENCE

CARD 04 / 12 • FRAC-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

PROPORTIONAL HEIGHT & 4-WAY MULTI-NOTATION

Fractions are proportional lengths on a 7mm grid.

Each block face shows fraction, decimal, and percent:

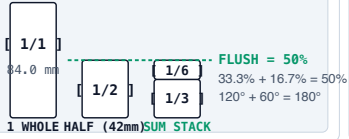
[1/3] (Symbol) = 0.333 (Decimal) = 33.3% (Percent)

Equal heights prove mathematical equivalence!

UNIVERSAL 84.00 MM WHOLE DATUM

- 1 Whole Datum: [1/1] = 12U = 84.00 mm reference rod
- Unit Base: [1/12] = 1U = 7.00 mm (Matches [2])
- Sub-Units: [1/2]=42mm, [1/3]=28mm, [1/4]=21mm

PROPORTIONAL EQUIVALENCE: 1/3 + 1/6 = 1/2



2. KIT PARTS INVENTORY (65 PIECES)

COMPONENT	QTY	SPEC	COLOR
Whole [1/1]	2x	84.00 mm	PLA Tan
Halves [1/2]	4x	42.00 mm	PLA Tan
Thirds [1/3]	3x	28.00 mm	Muted Red
Fourths [1/4]	4x	21.00 mm	Matte Blue
Fifths [1/5]	5x	16.80 mm	PLA White
Sixths [1/6]	6x	14.00 mm	Muted Red
Eighths [1/8]	8x	10.50 mm	Matte Blue
Ninths [1/9]	9x	9.33 mm	Muted Red
Tenths [1/10]	10x	8.40 mm	PLA White
Twelfths [1/12]	12x	7.00 mm	Matte Blue
13U Box & Lid	2x	98.4 mm	PLA Gray/White

TOTAL: 63 Blocks (12 Columns × 84mm) + 13U Box + Stage Lid = 65 Parts

SLANT 3D PRODUCTION PALETTE

- PLA White: Base-10 fractions (5x 1/5, 10x 1/10) & Stage Lid.
- PLA Matte Blue: Binary/duodecimal family (4x 1/4, 8x 1/8, 12x 1/12).
- PLA Matte Muted Red: Ternary fraction family (3x 1/3, 6x 1/6, 9x 1/9).
- PLA Tan: Unit wholes & halves (2x 1, 4x 1/2) • PLA Gray: 13U Box Body.

3. PACK-UP & SUPPORT

✓ **10-SECOND PACK-UP**

Stand all 63 blocks across the 12 floor sockets (2 cols of 1, 2 cols of 1/2, and 1 col each of 1/3..1/12). Every column levels at 84.0 mm; missing pieces leave an instant

✓ **NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ **HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Proportional height is strictly preserved under common denominator fraction addition.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: COMMON DENOMINATOR ADDITION & EQUIVALENCE

Step-by-Step Hands-On Tutorial: 1/3 + 1/4 = 7/12 & 4-Way Multi-Notation

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/fraction-studio/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 5.NF.A.1 (Unlike Denom...)
- CCSS 4.NF.A.2 (Fraction Com...)
- TEKS 111.7.b.3.H (Rational Ope...)

Grades 3–8 Fractions & Proportions

GUIDING INQUIRY

Which smaller block family levels off flush with both fraction towers in the comparison slot?

1 STEP 1: SEAT 1/1 WHOLE IN REFERENCE WELL

Place the 84.00 mm Whole [1/1] block in the left reference slot.
Its 4 faces show: 1/1, 1.00, 100%, and 360°.
This physical block establishes the 100% benchmark height.

2 STEP 2: STACK 1/3 AND 1/4 BLOCKS

Place one [1/3] block (28.00 mm) in the right comparison well.
Stack one [1/4] block (21.00 mm) directly on top of the [1/3].
Total combined height: 28.00 + 21.00 = 49.00 mm.

3–4 STEPS 3 & 4: TEST TWELFTHS & SOLVE

3. TEST COMMON TWELFTHS: Stack 7 twelfths: 7 × 7.00 mm = exactly 49.00 mm! Flush summit!
4. READ 4-WAY FACES: 0.333 + 0.250 = 0.583; 120° + 90° = 210°.
Calculator Verification: Press Fraction Display Icon ([FDISP] & [g][SIMP]) and run 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [+] [FDISP] → 7/12.

5 CORE PRINCIPLE: COMMON DENOMINATORS ARE COMMON SIZES

You cannot directly add thirds and fourths because their blocks are different physical heights!
Converting to twelfths (4/12 + 3/12 = 7/12) literally swaps disparate pieces for identical 7.00 mm unit blocks so their heights can be counted directly!

★ EXTENSION INVESTIGATION: ADD: 1/2 + 1/3

Seat [1/2] (42mm) and [1/3] (28mm) together. Test sixths: how many [1/6] blocks reach the exact 70.0mm summit? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.